Academic case study | Sophia Learning — Introduction to Networking
GreenLeaf Wireless Networking, Monitoring & Logs

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1. Wireless Network Design

1a. SSID Table

SSID Name	VLAN	Purpose	Security	Authentication	Band (5 GHz / 2.4 GHz)
GreenLeaf – Corporate	40	Employee corporate wireless access	WPA2-Enterprise	802.1X / RADIUS	5 GHz preferred, 2.4 GHz supported
GreenLeaf – Guest	50	Visitor/guest internet access	WPA2-Personal	Pre-Shared Key (PSK)	2.4 GHz only
GreenLeaf – Warehouse	50	Inventory scanner access	WPA2-Enterprise	802.1X / RADIUS	2.4 GHz only

1b. Wireless Standards and Radio Settings

GreenLeaf uses 802.11ac (Wi-Fi 5) on the 5 GHz band and 802.11n (Wi-Fi 4) on the 2.4 GHz band. The 5 GHz radios use 40 MHz channel widths, while the 2.4 GHz radios use 20 MHz channels to help reduce interference. At Portland, the APs use 17 dBm transmit power on 5 GHz and 14 dBm on 2.4 GHz. The Eugene office AP uses the same power levels, while the warehouse AP uses higher power at 20 dBm on 5 GHz and 17 dBm on 2.4 GHz to provide better coverage around the warehouse. The Portland Aironet 2802i APs also support 4x4 MIMO, MU-MIMO, and beamforming to improve wireless performance and efficiency.

1c. Channel Selection & AP Placement

GreenLeaf uses non-overlapping channels to reduce wireless interference. At Portland, the 5 GHz APs use channels 36, 149, and 48, while the 2.4 GHz APs use channels 1, 6, and 11. Channels 1, 6, and 11 are used because they do not overlap, and the 2.4 GHz channel width is kept at 20 MHz to further reduce interference. Portland has two APs on the first floor and one on the second floor, while Eugene has one AP in the office area and one mounted higher on the warehouse floor. Coverage is generally adequate, although the Eugene warehouse far end has weaker 5 GHz coverage of about -68 to -72 dBm, and an additional AP near the receiving dock may be needed if connectivity problems occur.

1d. Wireless Security Implementation

GreenLeaf-Corporate and GreenLeaf-Warehouse use WPA2-Enterprise with 802.1X/RADIUS authentication, while GreenLeaf-Guest uses WPA2-Personal with a pre-shared key (PSK). The guest network uses client isolation, a captive portal, a four-hour session timeout, and a 10 Mbps per-client bandwidth limit, while the corporate network uses band steering to prefer 5 GHz. The warehouse network is optimized for handheld scanners on 2.4 GHz and uses a DTIM period of 3 to help balance battery life and responsiveness.

2. Network Performance Monitoring

2a. Monitoring Tools and Platforms

GreenLeaf uses PRTG Network Monitor as its main monitoring platform and dashboard for tracking network health, availability, performance, and alerts. SNMP is used to monitor network devices such as routers and switches, while WMI collects performance information from Windows servers. NetFlow is used to analyze traffic patterns and bandwidth usage, and the PRTG dashboards bring this information together so administrators can quickly see device status, utilization, latency, and other performance issues.

2b. Baseline Performance Metrics

Segment	Baseline Metric	Value	Peak	Threshold
Portland WAN	Combined bandwidth	73 Mbps avg	245 Mbps	80% utilization
VPN Tunnel	Latency (RTT)	12 ms avg	25 ms	30 ms
PDX-SQL01	CPU usage	28% avg	65%	80%
VoIP	One-way latency	18 ms avg	N/A	150 ms

2c. QoS Summary

GreenLeaf uses QoS classes including VOICE, SIGNALING, BUSINESS-CRITICAL, BUSINESS-DATA, and DEFAULT to prioritize different types of network traffic. Voice traffic is marked DSCP EF (46) and receives priority bandwidth, while signaling uses CS3 (24) and business-critical traffic uses AF31 (26) with guaranteed bandwidth. VoIP performance is monitored using metrics such as MOS, with GreenLeaf maintaining an average MOS of 4.3 and a target above 4.0.

3. Logging & Alerts

3a. Syslog Architecture

GreenLeaf collects network logs on the central syslog server PDX-SYSLOG01 at 10.10.30.30 using rsyslog and Graylog. Logs are stored under /var/log/network/ and retained for 90 days, with facilities such as local0 for routers, local1 for switches, local2 for wireless access points, and local3 for firewalls. Devices sending logs include the Portland and Eugene routers, core and access switches, wireless APs, and the PDX-FW01 firewall.

3b. Normal Log Activity

During normal operations, the logs show routine DHCP activity, such as Portland-RTR assigning IP address 10.10.10.87 to a client at 08:12, followed by PDX-ACC-SW01 confirming the DHCP binding. The logs also show normal wireless activity, such as station 0050.5600.0055 associating with the GreenLeaf-Corporate SSID at 08:00. These events represent expected network behavior because devices regularity obtain IP addresses and connect to the wireless network during normal use.

3c. Anomalous Log Activity

The anomalous logs show several events that require investigation. Repeated SSH login failures from 198.51.100.200 indicate a brute-force attack, which triggered an authentication block after the failure limit was reached. On PDX-ACC-SW01, multiple unknown MAC addresses appeared on Fa0/24, causing port-security violations and suggesting that an unauthorized device may have been connected. PDX-CORE-SW01 also received a superior BPDU on Gi1/0/24, indicating an attempted STP attack, but Root Guard successfully blocked the port and protected the root bridge.

The logs also show the VPN tunnel between Portland and Eugene repeatedly going down and coming back up, which could disrupt communication between the sites. The VLAN 40 DHCP scope reached 95% utilization and clients began receiving NACKs because addresses were unavailable, creating a risk that wireless clients could not connect. Most critically, PDX-FILE01 ran suspicious svch0st.exe and mimikatz.exe processes,

connected to an unknown external IP, and created an unauthorized “hacker” account, indicating an active system compromise requiring immediate incident response.

3d. Defined Alerts

Authentication Failure: Triggered when more than five login failures come from the same source within 10 minutes. It is classified as a Warning, and an email is sent to the security-alerts team so the source can be investigated for a possible brute-force or credential attack.

VPN Tunnel Down: Triggered whenever a CRYPTO log reports that the VPN tunnel is down or deleted. It is classified as Critical and generates an immediate email and SMS notification so administrators can investigate and restore connectivity between the Portland and Eugene sites.

Port Security Violations: Triggered by any port-security violations message. It is classified as Critical and generates an immediate email notification so IT can investigate the affected switch port for an unauthorized or rogue device.